This video is copyrighted by M. Borahan Tümer, Marmara University and is protected under the Law on Intellectual and Artistic Works (No:5486) (T.C. 5486 no'lu Fikir ve Sanat Eserleri Kanunu) and all applicable international, federal, state and local laws, with all rights reserved. No part of this may be copied, distributed, or changed in any format, sold or used in any way other than individual learning purposes without express permission from M. Borahan Tümer, Marmara University.

1

1

Analysis & Design Document

A Guide for its Preparation…

2

2

Introduction

- ADD is a continuation of the project specification document (PSD)
- Goal: presenting the analysis and design of the project.
- Focuses on two different types of projects:
  - Software oriented projects
  - Academia oriented projects
- Type of project to determine by advisor !!!

3

3

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
- References

4

4

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
- References

5

5

Introduction

- Problem Description and Motivation (revised from PSD)
  - What is the *problem*?
  - Why is it *worth solving* it?
- Scope of the Project (revised from PSD)
  - What is *in* and what is *beyond* the project?
  - What to *use* and what to *develop*?
- Definitions, acronyms, and abbreviations

6

6

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
- References

7

7

Literature Survey

- Refine your literature survey in PSD with additional literature check by
  - adding any possibly new or just missed work related to your problem/work.
- Any new work critically similar to your project you should rigorously discuss in related work.

8

8

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
- References

9

9

Project Requirements

- *Functional* Requirements
  - *Description of system functionality*
    - System's reaction to particular inputs and
    - System's behaviour in particular situations.
- *Non-functional* Requirements
  - *Properties* of the system
  - *Constraints on the*
    - *services or functions* offered by the system such as timing constraints,
    - *development process, standards*, etc.
  - Must be ***verifiable*** (i.e., use a testable measure).

10

10

Non-Functional Requirements

- *Performance* *(throughput, respsonse time, screen refreshing time)*
- *Reliability* *(ROFOC, MTTF)* *(rate of failure occurrence, mean time to failure)*
- *Usability* *(Ease of use, # of help frames, warn./err. messages, size of user's guide)* *(# of errors made per unit time/training time)*
- *Security* *(for neworking systems; number of unauthorized accesses to the system)*
- *Maintainability* *(code modularity, rate of bug repair)()*
- *Portability/platform-independence (% target dependent statements)*
- *Scalability* *(# of running components, max. # of users with system up)*

11

11

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
- References

12

12

System Design

- *UML Use case diagrams* for main use cases
  - to specify *scenarios / use cases* of the system
- *UML Class and/or Database ER diagrams…*
  - to show building blocks of your system
  - to represent *a static view* of your model
  - describes attributes and behavior for targeted operations
- *UML Sequence diagrams…*
  - to describe sequence of *interactions among actors and use cases* to attain the goal of the specified behaviour.
- *User Interface (a preliminary version)*
- Test Plan

13

13

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
- References

14

14

Software Architecture

- Display
  - the system's *structure* and *static aspects* such as *subsystems/modules* using *block diagrams*, and other *visual aids*.
  - its *dynamic perspective* such as *data* and *control flow* using *state machines (especially fit for real-time systems)*

15

15

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
- References

16

16

Tasks Accomplished

- *Current state* of the project (implementation and preliminary results)
- *Task Log* (information about meetings and activities, including date, short description and hours)
- *Task Plan with Milestones* (clear and well-defined descriptions of the *work that must be completed* before predetermined check points, illustrated by *Gantt chart*)

17

17

Software-Oriented Project: Document Layout

- Title Page
- Introduction
- Literature Survey (revised from PSD)
- Project Requirements
- System Design
- Software Architecture (main aspects of data flow/control flow/modular design)
- Tasks Accomplished
- References

18

18

References

- *Use a specific numbering index with square brackets*
- *Add all resources* that you have made use of
- Do not add any resources you do not cite in the text!
- *Never misuse any source for copying/pasting* text! That is considered *plagiarism*!!! You should and would not do this!

19

19

Academic Oriented Project: Document Layout

- Title Page
- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

20

20

Academic Oriented Project: Document Layout

- Title Page
- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

21

21

Academic Oriented Project: Document Layout

- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- Preliminaries (NOT in ADD; only in your final book!!!)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

22

22

Introduction

- Problem Description and Motivation (revised from PSD)
  - What is the *problem*?
  - Why is it *worth solving* it?
- Scope of the Project (revised from PSD)
  - What is *in* and *beyond* the context of the project?
  - What to *use off-the-shelf* and to *develop*?
- Definitions, acronyms, and abbreviations

23

23

Academic Oriented Project: Document Layout

- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- Preliminaries (NOT in ADD; only in your final book!!!)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

24

24

Related Work

- *Refine* your *literature survey (PSD)* with *additional literature check* and *add* any possibly *new* or *just missed* work related to your problem/work.
- *Rigorously discuss* any newly found *work/method* essential/*comparable to your work* here.

25

25

Academic Oriented Project: Document Layout

- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- Preliminaries (NOT in ADD; only in your final book!!!)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

26

26

Preliminaries

- Provide, if any, necessary *fundamental or background concepts* you need know to accomplish your main tasks in this project.
- Be rigorous in including all necessary *fundamentals* etc. you will be exploiting.

27

27

Academic Oriented Project: Document Layout

- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- Preliminaries (NOT in ADD; only in your final book!!!)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

28

28

System Design (Methodology)

- *System Model (block diagrams for the static and state machines for dynamic perspective, other visual aids welcome)*
- *Flow charts and/or pseudocode of proposed algorithms*
- *Performance assessment criteria, comparison metrics (provide detailed explanation, the more unambiguous the better!!!)*
- *Data sets or benchmarks (provide detailed explanation)*

29

29

System Design (Methodology)

- *Flow charts and/or pseudocode of proposed algorithms*
  - Give enough details!
  - Also consider a separate pseudocode per module!
- *Performance assessment criteria, comparison metrics*
  - select one of (Adjusted Rand Index (ARI), F(1)-score, precision/recall or other standard accuracy criteria such as sensitivity, specificity, false positive/negative rates, etc.) with your advisor suitable to compare your results with those of the baseline algorithms in the literature.

30

30

Academic Oriented Project: Document Layout

- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- Preliminaries (NOT in ADD; only in your final book!!!)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

31

31

Software Architecture

- Display
  - the system's *structure* and *static aspects* such as *subsystems/modules* using *block diagrams*, and other *visual aids*.
  - its *dynamic perspective* such as *data* and *control flow* using *state machines (especially fit for real-time systems)*

32

32

Academic Oriented Project: Document Layout

- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- Preliminaries (NOT in ADD; only in your final book!!!)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

33

33

Experimental Study

- *Experiment setup*
  - For each experiment that you conduct, specify
    - what *idea(s)* this experiment *support*
    - *how it works*
    - any *parameter values, necessary system specs*
- *Results*
  - *show* your results *using curves, tables, figures and other visual aids*, etc
- *Detailed discussion*
  - Study and clarify *all aspects of the results*.

34

34

Academic Oriented Project: Document Layout

- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- Preliminaries (NOT in ADD; only in your final book!!!)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

35

35

Tasks Accomplished

- *Current state* of the project (implementation and preliminary results)
- *Task Log* (information about meetings and activities, including date, short description and hours)
- *Task Plan with Milestones* (clear and well-expressed descriptions of the *work that must be completed* before predetermined check points, illustrated by *Gantt chart*)

36

36

Academic Oriented Project: Document Layout

- Introduction
- Related Work (Comprehensive literature survey to present state-of-the art methods/algorithms)
- Preliminaries (NOT in ADD; only in your final book!!!)
- System Design
- System Architecture (main aspects of data flow/control flow)
- Experimental Study (draft version to summarize details of the experiments)
- Tasks Accomplished
- References

37

37

References

- *Use a specific numbering index with square brackets*
- *Add all resources* that you have made use of
- Do not add any resources you do not cite in the text!
- *Never misuse any source for copying/pasting* text! That is considered *plagiarism*!!! You should and would not do this!

38

38